JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY

Scheme of Teaching and Examination 2024-25

Outcome Based Education (OBE) and Choice Based Credit System (CBCS)

(Effective from the academic year 2024-25)

SCHEME OF TEACHING AND ASSESSMENT FOR B.E

Semester I: Physics cycle

Sl. No.	Course code	Course Title	Category code	Teaching Department	QP Setting Dept.	Teaching Hours per Week				Credits	Marks			Exam Duration in hours
						L	T	P	Total		CIE	SEE	Total	
1	24MA110	Engineering Mathematics – I	BSC	MA	MA	3	1	-	4	4	40	60	100	03
2	24PH110	Engineering Physics	BSC	PH	PH	3	-	2	5	4	40	60	100	03
3	24EE110	Elements of Electrical Engineering	ESC	EE	EE	3	-	-	3	3	40	60	100	03
4	24ME11x	Elective 1 – Mechanical Engineering	ESC	ME	ME	3	-	-	3	3	40	60	100	03
5	24CS11x	Elective 2 - Introduction to Programming	ESC	CS/IS	CS/IS	3	-	2	5	4	40	60	100	03
6	24HU110	Kannada	HSMC	HU (KA)	HU (KA)	2	-	-	2	0	40	-	40	-
7	24HU120	Innovation & Design Thinking	HSMC	PD	HU	2	-	-	2	1	20	30	50	02
8		NSS/NCC/SDA								-	-	-	-	-
Total										**19**	**260**	**330**	**590**	-

Elective 1 – Mechanical Engineering	
24ME111	Elements of Mechanical Engineering
24ME112	Mechatronics and Robotics

Elective 2 - Introduction to Programming	
24CS111	C Programming
24CS112	Python Programming
24CS113	Object Oriented Programming using JAVA

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

Semester I: Chemistry cycle

SlNo	Course code	Course Title	Category code	Teaching Dept.	QP Setting Dept.	Teaching Hours per Week				Credits	Marks			Exam Duration in hours
						L	T	P	Total		CIE	SEE	Total	
1	24MA110	Engineering Mathematics –I	BSC	MA	MA	3	1	-	4	4	40	60	100	03
2	24CH110	Engineering Chemistry	BSC	CH	CH	3	-	2	5	4	40	60	100	03
3	24CV110	Engineering Mechanics	ESC	CV	CV	3	-	-	3	3	40	60	100	03
4	24EC11x/ 24EI11x	Elective 3 – Electronics Engineering	ESC	EC/EI	EC/EI	3	-	-	3	3	40	60	100	03
5	24ME120	Engineering Graphics and Design	ESC	ME	ME	-	1	2	3	2	40	60	100	03
6	24HU11x	Elective 4 – Sustainability/Skill	HSMC	All Dept	HU	2	-	-	2	2	20	30	50	02
7	24HU130	Communication English	HSMC	HU (EN)	HU (EN)	2	-	-	2	1	20	30	50	02
8		NSS/NCC/SDA/Sports/Yoga								-	-	-	-	-
Total										**19**	**240**	**360**	**600**	-

Elective 3 – Electronics Engineering	
24EC111	Elements of Electronics Engineering
24EC112	Fundamentals of Robotics and its Applications
24EC113	Sensors and Measurements
24EC114	Internet of Everything
24EI115	Transducers and Measuring Systems

Elective 4 – Sustainability/Skill	
24HU111	Sustainability Engineering
24HU112	Introduction to Cyber Security
24HU113	Human Psychology
24HU114	Web Technology
24HU115	Biology for Engineers
24HU116	Carbon Sequestration

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

Semester II: Physics cycle

SlNo	Course code	Course Title	Category code	Teaching Department	QP Setting Dept.	Teaching Hours per Week				Credits	Marks			Exam Duration in hours
						L	T	P	Total		CIE	SEE	Total	
1	24MA210	Engineering Mathematics – II	BSC	MA	MA	3	1	-	4	4	40	60	100	03
2	24CH210	Engineering Chemistry	BSC	CH	CH	3	-	2	5	4	40	60	100	03
3	24CV210	Engineering Mechanics	ESC	CV	CV	3	-	-	3	3	40	60	100	03
4	24EC21x 24EI21x	Elective 3 – Electronics Engineering	ESC	EC/EI	EC/EI	3	-	-	3	3	40	60	100	03
5	24ME220	Engineering Graphics and Design	ESC	ME	ME	-	1	2	3	2	40	60	100	03
6	24HU21x	Elective 4 – Sustainability/Skill	HSMC	All Dept	HU	2	-	-	2	2	20	30	50	02
7	24HU230	Communication English	HSMC	HU (EN)	HU (EN)	2	-	-	2	1	20	30	50	02
8		NSS/NCC/SDA/Sports/Yoga								-	-	-	-	-
Total										**19**	**240**	**360**	**600**	

Elective 3 – Electronics Engineering	
24EC111	Elements of Electronics Engineering
24EC112	Fundamentals of Robotics and its Applications
24EC113	Sensors and Measurements
24EC114	Internet of Everything
24EI115	Transducers and Measuring Systems

Elective 4 – Sustainability/Skill	
24HU211	Sustainability Engineering
24HU212	Introduction to Cyber Security
24HU213	Human Psychology
24HU214	Web Technology
24HU215	Biology for Engineers
24HU216	Carbon Sequestration

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

Semester II: Chemistry cycle

Sl. No.	Course code	Course Title	Category code	Teaching Department	QP Setting Dept.	Teaching Hours per Week				Credits	Marks			Exam Duration in hours
						L	T	P	Total		CIE	SEE	Total	
1	24MA210	Engineering Mathematics – II	BSC	MA	MA	3	1	-	4	4	40	60	100	03
2	24PH210	Engineering Physics	BSC	PH	PH	3	-	2	5	4	40	60	100	03
3	24ME21x	Elective 1 – Mechanical Engineering	ESC	ME	ME	3	-	-	3	3	40	60	100	03
4	24EE210	Elements of Electrical Engineering	ESC	EE	EE	3	-	-	3	3	40	60	100	03
5	24CS21x	Elective 2 – Introduction to Programming	ESC	CS/IS	CS/IS	3	-	2	5	4	40	60	100	03
6	24HU210	Kannada	HSMC	HU (KA)	HU (KA)	2	-	-	2	0	40	-	40	-
7	24HU220	Innovation & Design Thinking	ESC	PD	HU	2	-	-	2	1	20	30	50	02
8		NSS/NCC/SDA/Sports/Yoga								-	-	-	-	-
Total										**19**	**260**	**330**	**590**	-

Elective 1 – Mechanical Engineering	
24ME211	Elements of Mechanical Engineering
24ME212	Mechatronics and Robotics

Elective 2 - Introduction to Programming	
24CS211	C Programming
24CS212	Python Programming
24CS213	Object Oriented Programming using JAVA

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

Semester III:

Sl. No.	Course Code	Course Title	Category Code	Teaching Department	QP Setting Dept.	Teaching Hours per Week				Credits	Marks			Exam Duration in hours
						L	T	P	Total		CIE	SEE	Total	
1	24MATDIP310	Foundations of Engineering Mathematics-I	BSC	MA	MA	3	0	0	3	-	-	60	60	03
2	24MA313	Engineering Mathematics - III	BSC	MA	MA	3	0	0	3	3	40	60	100	03
3	24CS310	Object Oriented Programming using C++	PCC	CS	CS	3	0	2	5	4	40	60	100	03
4	24CS320	Digital System Design	PCC	CS	CS	3	0	2	5	4	40	60	100	03
5	24CS330	Data Structures	PCC	CS	CS	4	0	0	4	4	40	60	100	03
6	24CS340	Computer Organization	PCC	CS	CS	3	0	0	3	3	40	60	100	03
7	24CS350	Discrete Mathematical Structures	PCC	CS	CS	3	0	0	3	3	40	60	100	03
8	24CS36L	Data Structures Lab	PCCL	CS	CS	0	0	3	3	1.5	40	60	100	03
9	24HU311	Universal Human Values-II	HSMC	CS	HU	2	0	0	2	2	20	30	50	02
		NSS/NCC/SD/Sports/ Yoga	SDA							-	-	-	-	-
Total										**24.5**	**300**	**450/ 510***	**750/ 810***	

*For lateral entry student's total marks is 810

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

Semester IV:

Sl. No.	Course Code	Course Title	Category Code	Teaching Department	QP Setting Dept.	Teaching Hours per Week				Credits	Marks			Exam Duration in hours
						L	T	P	Total		CIE	SEE	Total	
1	24MATDIP410	Foundations of Engineering Mathematics-II	BSC	MA	MA	3	0	0	3	0	-	60	60	03
2	24MA413	Linear Algebra	BSC	MA	MA	3	0	0	3	3	40	60	100	03
3	24CS410	Operating Systems	PCC	CS	CS	3	0	2	5	4	40	60	100	03
4	24CS420	Design and Analysis of Algorithms	PCC	CS	CS	4	0	0	4	4	40	60	100	03
5	24CS430	Data Communication	PCC	CS	CS	4	0	0	4	4	40	60	100	03
6	24CS440	Software Engineering	PCC	CS	CS	3	0	0	3	3	40	60	100	03
7	24CS45xOE	Open Elective-I	OEC	CS	CS	3	0	0	3	3	40	60	100	03
8	24CS46L	Design and Analysis of Algorithms Lab	PCCL	CS	CS	0	0	3	3	1.5	40	60	100	03
9	24HU412	Environmental Studies	HSMC	EV	EV	2	0	0	2	2	20	30	50	02
		NSS/NCC/SD/Sports/Yoga	SDA							-	-	-	-	-
Total										**24.5**	**300**	**450/ 510***	**750/ 810***	

*For lateral entry student's total marks is 810

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

Semester V:

Sl. No.	Course Code	Course Title	Category Code	Teaching Department	QP Setting Dept.	Teaching Hours per Week				Credits	Marks			Exam Duration in hours
						L	T	P	TOTAL		CIE	SEE	Total	
1	24CS510	Computer Networks	PCC	CS	CS	3	0	2	5	4	40	60	100	03
2	24CS520	Database Management Systems	PCC	CS	CS	3	0	0	3	3	40	60	100	03
3	24CS530	Theory of Computation	PCC	CS	CS	3	0	0	3	3	40	60	100	03
4	24CS540	Artificial Intelligence	PCC	CS	CS	3	0	0	3	3	40	60	100	03
5	24CS55x	Professional Elective-I	PEC	CS	CS	3	0	0	3	3	40	60	100	03
6	24CS56x	Professional Elective-II	PEC	CS	CS	3	0	0	3	3	40	60	100	03
7	24CS57xOE	Open Elective-II	OEC	CS	CS	3	0	0	3	3	40	60	100	03
8	24CS58L	Database Management Systems Lab	PCCL	CS	CS	0	0	3	3	1.5	40	60	100	03
		NSS/NCC/SD/Sports/Yoga	SDA							-	-	-	-	-
Total										**23.5**	**320**	**480**	**840**	

*Details provided in the syllabus

Professional Elective-I		Professional Elective-II	
24CS551	Image Processing and Computer Vision	24CS561	JAVA and J2EE
24CS552	Data Science and Applications	24CS562	Computer Architecture
24CS553	Multimedia Computing	24CS563	Software Testing and Quality Assurance

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

Semester VI:

Sl. No.	Course code	Course title	Category Code	Teaching Department	QP Setting Dept.	Teaching Hours per Week				Credits	Marks			SEE Duration in hours
						L	T	P	TOTAL		CIE	SEE	Total	
1	24CS610	Internet of Things	PCC	CS	CS	3	0	2	5	4	40	60	100	03
2	24CS620	Machine Learning	PCC	CS	CS	3	0	0	3	3	40	60	100	03
3	24CS630	Principles of Compilers	PCC	CS	CS	3	0	0	3	3	40	60	100	03
4	24CS640	Cryptography and Network Security	PCC	CS	CS	3	0	0	3	3	40	60	100	03
5	24CS65x	Professional Elective-III	PEC	CS	CS	3	0	0	3	3	40	60	100	03
6	24CS66x	Professional Elective-IV	PEC	CS	CS	3	0	0	3	3	40	60	100	03
7	24CS67xOE	Open Elective-III	OEC	CS	CS	3	0	0	3	3	40	60	100	03
8	24CS68L	Machine Learning Lab	PCCL	CS	CS	0	0	3	3	1.5	40	60	100	03
9	24HU610	Constitution of India and Professional Ethics	HSMC	HU	-	2	0	0	2	0	40	-	40	-
		NSS/NCC/SD/Sports/Yoga	SDA							-	-	-	-	-
Total										**23.5**	**360**	**480**	**840**	

*Details provided in the syllabus

Professional Elective-III		**Professional Elective-IV**	
24CS651	Cloud Computing	24CS661	Blockchain Technology
24CS652	Automotive Cyber Security	24CS662	Natural Language Processing
24CS653	Cyber Security	24CS663	Full Stack Development
24CS654	Data Mining and Warehousing	24CS664	Robotics

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

Semester VII:

Sl. No.	Course code	Course title	Category Code	Teaching Department	QP Setting Dept.	Teaching Hours per Week				Credits	Marks			SEE Duration in hours
						L	T/ SDA	P	TOTAL		CIE	SEE	Total	
1	24CS710	Software Project Management	HSMC	CS	CS	3	0	0	3	3	40	60	100	03
2	24CS72x	Professional Elective-V	PEC	CS	CS	3	0	0	3	3	40	60	100	03
3	24CS73x	Professional Elective-VI	PEC	CS	CS	3	0	0	3	3	40	60	100	03
4	24CS74P	Industrial training /Internship	PWC	CS	-	-	-	-	-	2	40	-	40	-
5	24CS75P	Project Work Phase – I	PWC	-	-	-	-	-	-	2	40	-	40	-
Total										**13**	**200**	**180**	**380**	

*Details provided in the syllabus

Professional Elective-V		**Professional Elective-VI**	
24CS721	Deep Learning	24CS731	Object Oriented Modeling and Design
24CS722	DevOps	24CS732	Text and Web Mining
24CS723	Big Data Analytics	24CS733	Enterprise Resource Planning

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SCHEME OF TEACHING AND ASSESSMENT FOR B.E

Semester VIII:

Sl. No.	Subject Code	Course title	Category Code	Teaching Department	QP Setting Dept.	Teaching Hours per Week				Credits	Marks			SEE Duration in hours
						L	T	P	TOTAL		CIE	SEE	Total	
1	24CS81x	Professional Elective-VII	PEC	CS	CS	3	0	0	3	3	40	60	100	03
2	24CS82P	Project work Phase – II	PWC	CS	CS	-	-	-	-	10	40	60	100	03
Total										**13**	**80**	**120**	**200**	

*Details provided in the syllabus

Professional Elective-VII	
24CS811	Game Theory
24CS812	High Performance Computing
24CS813	Quantum Computing